Screen-Based vs. Non-Screen Leisure: How Social Context Shapes Momentary Happiness and Stress

WELLESLEY

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Introduction

- Leisure activities occupy a substantial portion of daily life
- Screen-based leisure (television, computer use, games) may differ from non-screen leisure (reading, hobbies, relaxing) in its effects on well-being
- Social context, being alone versus with others, could change how an activity affects happiness and stress
- The ATUS Well-Being Module (2010, 2012, 2013) randomly selected three activities per respondent and measured momentary happiness and stress on a 0-6 scale.
- Prior work has found associations between screen time and lower psychological well-being (Twenge et al., 2018). Folk and Dunn (2026) found that social interaction is associated with higher happiness across daily activities

Research Question

How do momentary happiness and stress differ between screen-based and non-screen leisure activities, and do those patterns change depending on whether people are alone or with others?

Data

- Used the American Time Use Survey Well-Being Module from 2010, 2012, and 2013.
- The final dataset was merged at the episode level using the activity, well-being, respondent, roster, and Who files.
- The final analytic sample includes 12,444 respondents and 14,986 leisure episodes.

Variables / Sample Description

Key Variables

- **Outcomes:** Momentary happiness, momentary stress
- **Activity Type:**
  - Screen-based leisure: TV watching, leisure computer use, video games
  - Non-screen leisure: Reading, arts & crafts, relaxing
- **Social context**: Alone vs. with others
- **Controls:** Age, sex, employment status
- Excluded: Episodes with invalid well-being values; non-leisure episodes

Sample Description

- Screen-based episodes outnumber non-screen episodes: 11,273 vs. 3,713
- Largest group: Screen-based with others (n = 8,138); Smallest: Non-screen alone (n = 1,262)
- Mean age: 50.3 years (range 15-85); roughly equal sex split
- 81% of respondents contribute 1 episode; max 3 episodes per person (32.7% of episodes come from repeated respondents)

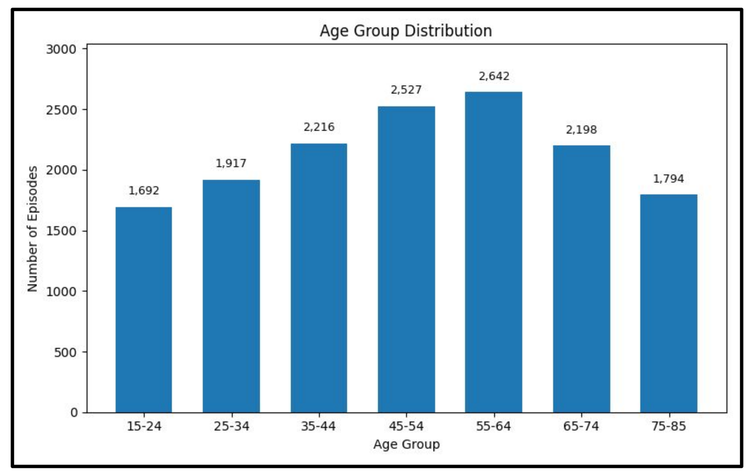

Figure 1. The analytic sample included episodes across all age groups, with the most episodes from ages 45–64.

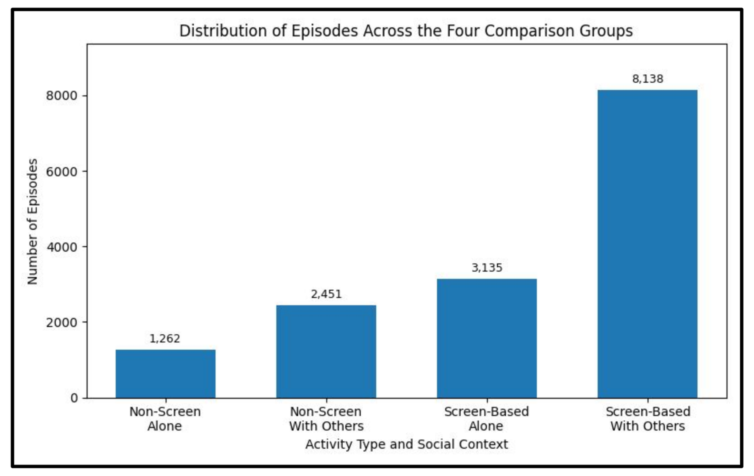

Figure 2. Screen-based leisure with others was the largest group in the sample, while non-screen leisure alone was the smallest.

Methods

Data Pipeline:

- Merged 5 ATUS files; constructed 6-digit activity codes; derived alone/with-others variable

Model 1: Weighted Least Squares (WLS) + Clustered Standard Errors (SEs)

- Applied ATUS activity weights for population-level estimates
- Clustered standard errors by respondent ID to account for repeated episodes

Model 2: Linear Mixed-Effects Model

- Random intercept per respondent to account for repeated episodes and baseline differences between people
- Tradeoff: mixed model was unweighted due to software limitations.

Model 3: Unweighted + Clustered Standard Errors

- Removed survey weights but still clustered standard errors by respondent ID
- Used as a sensitivity model to separate the role of survey weights from the role of model choice.

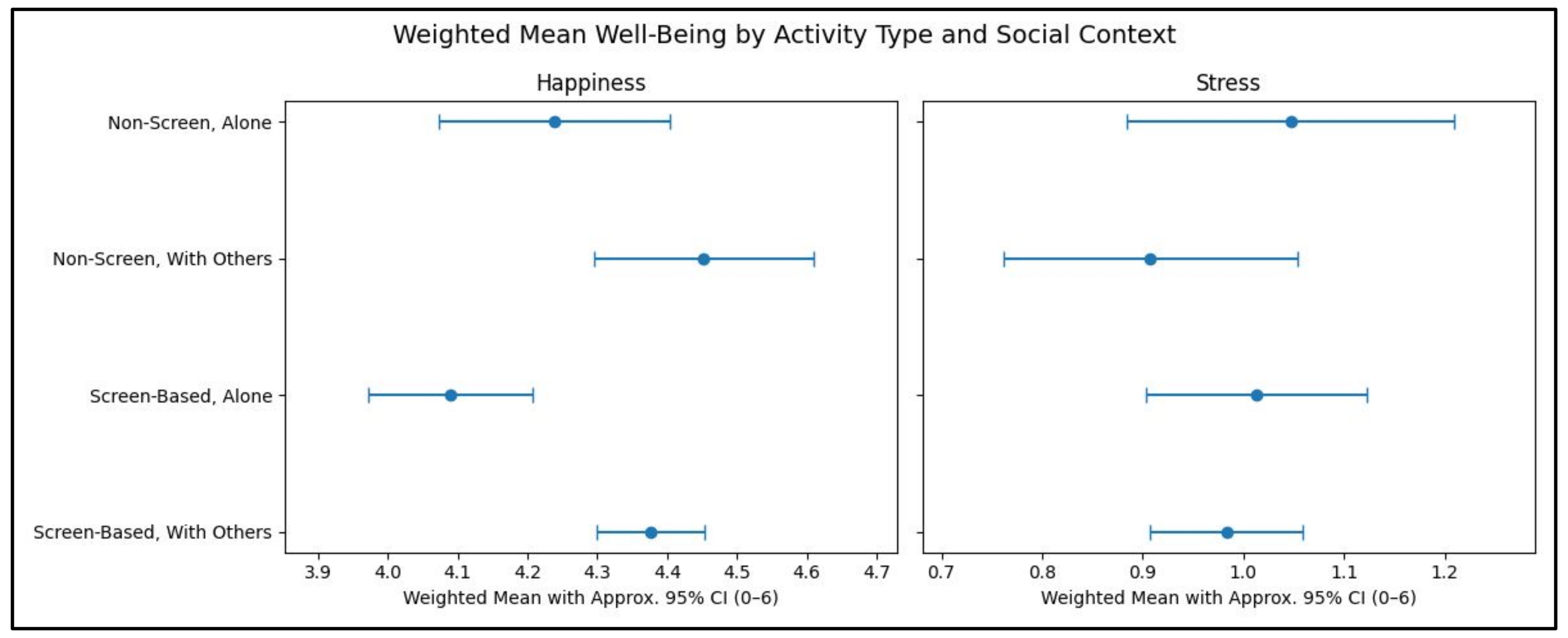

Figure 3. Points show weighted group means, and horizontal bars show approximate 95% confidence intervals around those means. Group differences were clearer for happiness than for stress: non-screen leisure with others had the highest happiness, while screen-based leisure alone had the lowest.

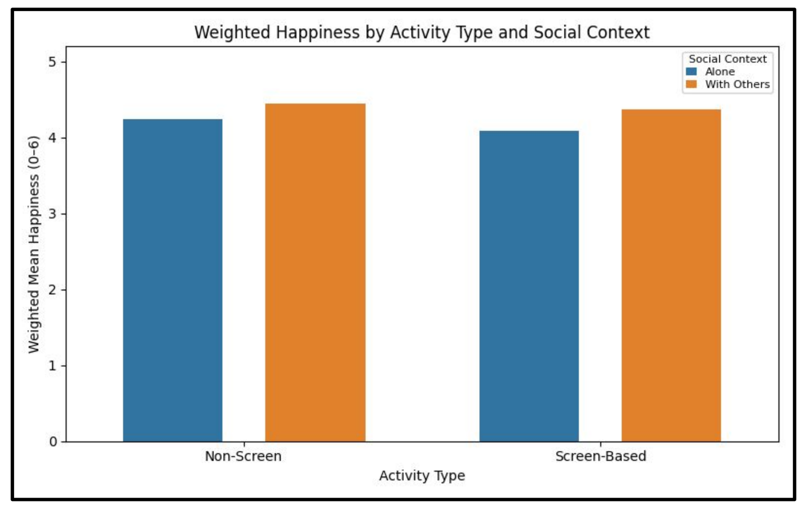

Figure 4. Weighted happiness was highest for non-screen leisure with others and lowest for screen-based leisure alone.

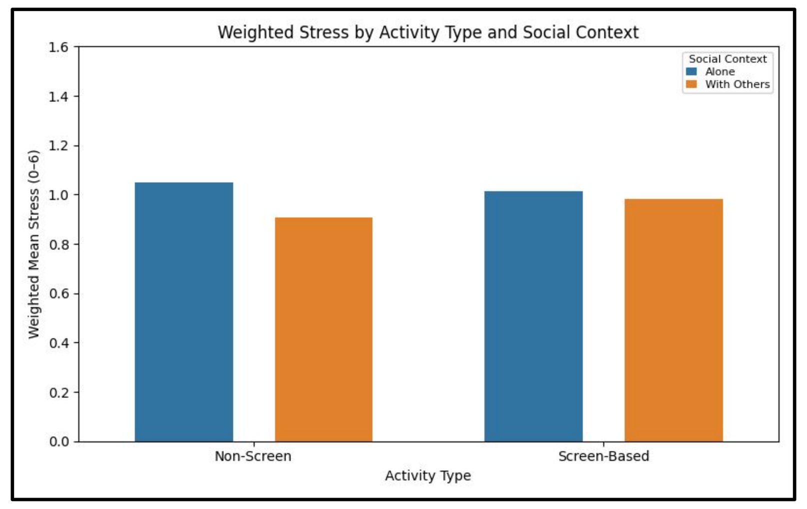

Figure 5. Stress differed less across groups than happiness. Non-screen leisure with others had the lowest weighted stress, while non-screen leisure alone had the highest.

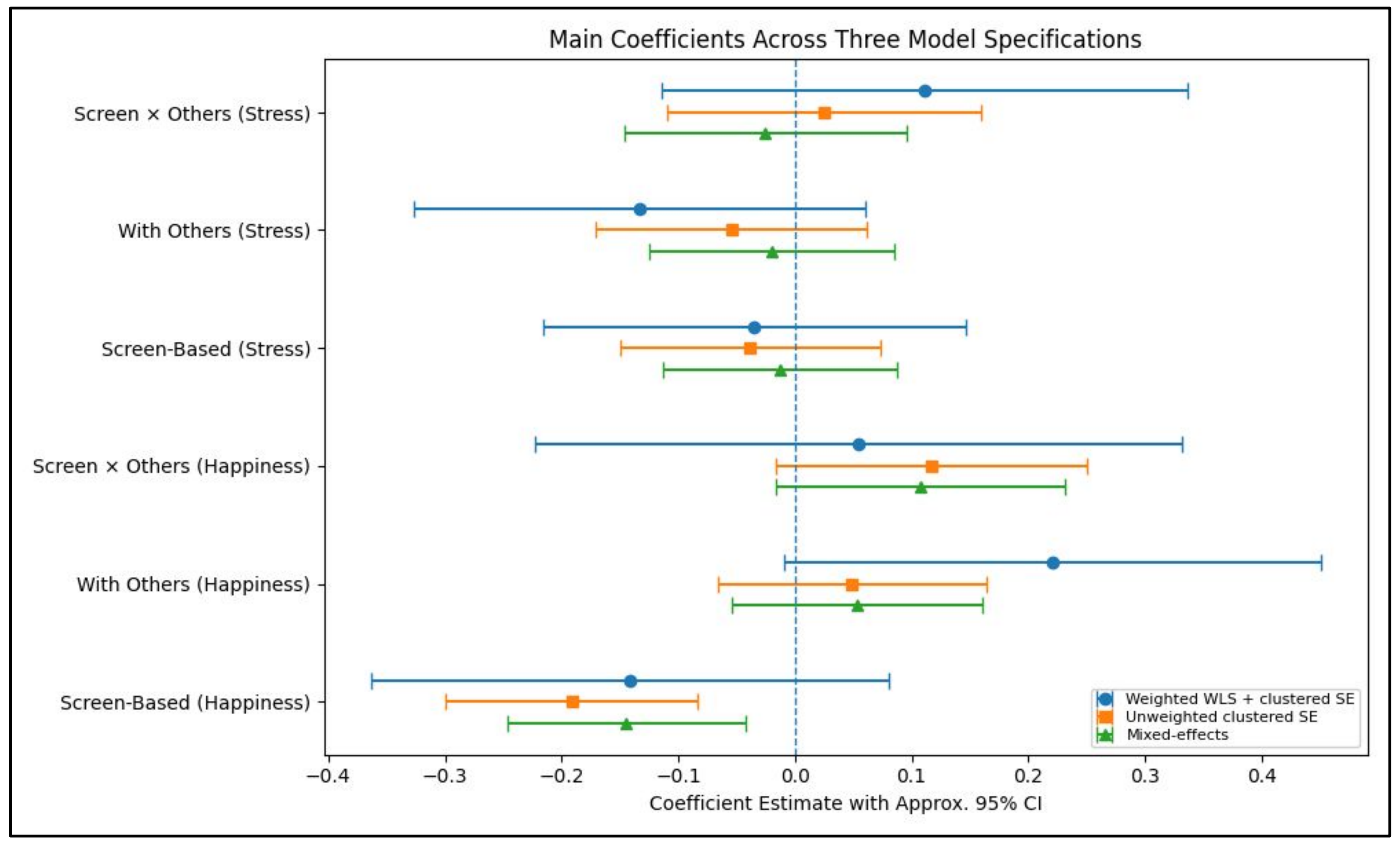

Figure 6. Coefficients were generally clearer for happiness than for stress, and all three models showed a similar negative screen-based happiness pattern.

Outcome	Term	Weighted WLS + clustered SE	Mixed-effects	Unweighted clustered SE
Happiness	Screen-Based	-0.141 (p=.212)	-0.144** (p=.006)	-0.191** (p=.001)
	With Others	+0.221 (p = .060)	+0.054 (p = .327)	+0.049 (p=.406)
	Screen × Others	+0.054 (p=.701)	+0.108 (p=.087)	+0.117 (p=.085)
	Fit	R^2=.010; Adj. R^2=.009	Marg. R^2=.004; Cond. R^2=.575	R^2=.005; Adj. R^2=.004
Stress	Screen-Based	-0.035 (p=.708)	-0.013 (p=.800)	-0.038 (p=.501)
	With Others	-0.133 (p=.177)	-0.020 (p=.708)	-0.054 (p=.361)
	Screen × Others	+0.112 (p=.332)	-0.025 (p=.682)	+0.026 (p=.710)
	Fit	R^2=.008; Adj. R^2=.007	Marg. R^2=.004; Cond. R^2=.636	R^2=.005; Adj. R^2=.004

Table 1. Model results. Coefficients are relative to the reference group: non-screen leisure, alone, male, employed full-time, with age included as a continuous control.

Results and Discussion

Happiness Findings:

- In the mixed-effects model, screen-based leisure was associated with 0.144 points lower happiness than non-screen leisure on the 0–6 scale, about a 2.4% difference (p = .006).
- The weighted WLS model showed a similar negative coefficient (β = -0.141), but it was not statistically significant. The unweighted clustered-SE model was stronger and significant (β = -0.191, p = .001), closer to the mixed-effects result. This points to survey weighting, rather than clustered SE versus mixed-effects structure, as the main source of difference.
- Being with others was marginally significant in WLS (β = 0.221, p = 0.060), but smaller and not significant in the mixed model (β = 0.054, p = 0.327).
- The Screen × Others interaction was not significant in either model, meaning social context did not significantly change the happiness gap between screen-based and non-screen leisure.

Stress Findings:

- None of the main activity or social-context predictors were statistically significant for momentary stress. Their 95% confidence intervals crossed zero.

Controls

- Being out of the labor force was significantly associated with lower happiness in all three models (weighted WLS: -0.208; unweighted: -0.130; mixed: -0.122) and higher stress in all three models (weighted WLS: +0.230; unweighted: +0.188; mixed: +0.168).
- Unemployment was significantly associated with higher stress across models (weighted WLS: +0.347; unweighted: +0.289; mixed: +0.308) and lower happiness in the unweighted and mixed models (-0.136).
- Compared to males, female respondents reported significantly higher happiness and stress in the unweighted and mixed models, while older age was significantly associated with slightly lower stress in those models.

Model Fit

- WLS R^2 and adjusted R^2 were nearly identical, showing no obvious overfitting.
- Cross-validation R^2 values were near zero across all models, showing limited predictive power, while similar RMSE values across folds suggested stable model performance.
- RMSE was about 1.66 points on the 0-6 scale across models.
- Marginal R^2 was low (0.004), but conditional R^2 was much higher (0.575–0.636), indicating that stable person-level differences explain far more variation than activity type or social context.
- VIF values were below 3.1, indicating no serious multicollinearity among controls.

Model	Training R^2	CV R^2(mean)	CV RMSE (mean±SD)
Mixed Happiness	0.004 (marg.)	0.003	1.657 ± 0.027
Mixed Stress	0.004 (marg.)	0.002	1.663 ± 0.029
Weighted WLS Happiness	.010	-0.001	1.660 ± 0.027
Weighted WLS Stress	0.008	-0.002	1.667 ± 0.032
Unweighted Clustered Happiness	0.005	0.003	1.657 ± 0.027
Unweighted Clustered Stress	0.005	0.002	1.663 ± 0.029

Table 2. Model fit and 5-fold cross-validation results. Training R^2 for mixed-effects models refers to marginal R^2, meaning fixed effects only. CV R^2 and RMSE are based on held-out respondents. RMSE is reported in the original 0-6 well-being scale.

Conclusion

- Screen-based leisure was consistently associated with lower momentary happiness, but the weighted model made this estimate less precise.
- The unweighted clustered-SE model was closer to the mixed-effects result, pointing to survey weights as the main source of the model difference.
- Social context did not significantly change the relationship between activity type and happiness.
- Stress was not meaningfully related to leisure type or social context.
- Low marginal R^2 with high conditional R^2 indicates that stable person-level differences explain much more variation than activity type or social context.

Limitations

- The analysis is observational; no causal claims
- Well-being ratings capture sampled episodes from one diary day, not long-term leisure patterns
- Social context collapsed to alone vs. with others; relationship quality not captured
- Activity groupings involve judgment calls; different groupings may yield different results
- A methodological tradeoff remains: weighted WLS supports population-level estimates, while mixed-effects models account more directly for repeated respondents. The unweighted clustered-SE check helps separate the role of survey weights from the role of model choice.

References

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